

STA 2243: FUNDAMENTALS OF COMPUTER INTERACTIVE STATISTICS

Lecture Notes
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Course Outline

Introduction to R and Excel: Numeric and vector operations and importing data from spreadsheets like excel and SPSS. Measures of central tendency: Arithmetic mean, Harmonic mean, Geometric Mean, Mode, Median, Quartiles. Measures of dispersion: Range, Variance, Kurtosis, Skewness, Exploratory data analysis. Correlation: Pearson and Rank correlations. Simple Linear Regression, Computation of density, distribution and quintile functions for hyper geometric, binomial, Poisson, normal, exponential, Gompertz weibull, uniform, and Chi-square, beta and gamma distributions. S-plus/R will be used throughout.

Introduction to R

RStudio is an IDE that provides features to make using and managing R much easier. Let's take a look at base R first.

Base R

To work with base R you have access to two ways to write and run your code.

1. The R console: this is the "interpreter" and runs your code in real time
2. R scripts: An R analysis script allows you to store your code in a static document that you can save. A script allows you to:
 - Save your code and to share with others (reproducibility)
 - Try things out interactively and then add/modify to your code in the script

Features of R

1. R is free
2. It runs on all major platforms
3. Scripts and data objects can be shared
4. It is easy for others to contribute add-ons which enables developers to share software implementation

RStudio IDE

RStudio is an integrated development environment (IDE) for R. It helps the user effectively use R by making things easier.

RStudio makes working with R easier by including features such as syntax highlighting, code completion, and smart indentation

Windows in RStudio

R Console

You can also type things here interactively.

Code is not saved on your disk when entered into the console

R Script

This will appear in top left corner of RStudio.

Environment

- Workspace/environment tab which tells you what objects are in R and what exists in memory/what is loaded/what you have read in.
- History tab which shows previous commands you have run.

Fourth Pane

- Files - shows the files on your computer in the directory you are working in
- Viewer - can view data or R objects
- Help - shows help documentations for R commands
- Plots - shows plots generated in your R sessions.
- Packages - list of R packages you have installed

R Projects and Intro to RMarkdown

R projects are a great way to manage all the files used in your R session.

Creating an R project can be extremely helpful in organizing your work:

- Helps you organize your work.
- Helps with working directories
- Allows you to easily know which project you're on.
- Allows a project folder to be zipped and sent to another R users who can unzip and pick up right where you left off.
- To create a new Rproject go to File > New Project > New Directory > New Project.

R Markdown

Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents.

The way you can create a file like this in RStudio is: File → New File → R Markdown and then using the default or using a template.

When you click the Knit button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document.

Basic R

In this lecture we will begin our introduction to R programming where we will execute:

1. R scripts and comments
2. R variables
3. Vectors

R as a calculator

R can be used as a basic calculator to do the following:

Addition

For addition you use

```
# addition
```

```
2+ 3
```

```
[1] 5
```

Subtraction

```
10-8
```

```
[1] 2
```

Multiplication

```
5*6
```

```
[1] 30
```

Division

```
100/2
```

```
[1] 50
```

Squareroot

```
sqrt(9)
```

```
[1] 3
```

Square/Powers

```
4^2
```

```
[1] 16
```

Sum

```
sum(c(1,2,3))
```

```
[1] 6
```

Vectors in R

A vector is the most common and basic data type in R, and is pretty much the workhorse of R.

A vector is composed by a series of values, which can be either numbers or characters.

We can assign a series of values to a vector using the `c()` function, which stands for concatenate (combine/connect one after another) values into a vector.

Example 1

We can create a vector of ages of students in class and assign them the object name `age` as follows

```
age <- c(18,19,20,18,23,25,27)
```

```
age
```

```
[1] 18 19 20 18 23 25 27
```

A vector also contains characters as shown below by a list of student names

```
names_student <- c("Anne", "Mark", "Job", "Stacy", "Samuel")

names_student
```

```
[1] "Anne" "Mark" "Job" "Stacy" "Samuel"
```

length() tells you how many elements are in a particular vector:

```
length(age) # contents in age vector
```

```
[1] 7
```

```
length(names_student) # contents in the names vector
```

```
[1] 5
```

An important feature of a vector is that all of the elements are the same type of data. The function class() indicates the class (the type of element) of an object:

```
class(age)
```

```
[1] "numeric"
```

```
class(names_student)
```

```
[1] "character"
```

One can create vectors using the seq() function in R.

The specification for the seq() function is:

1. from: starting values of the sequence
2. to: End values of the sequence
3. by: increment of the sequence
4. length.out: desired length of the sequence

Example we create a sequence of numbers between 2 and 6 increasing by 0.5

```
x <- seq(from = 2, to = 6, by = 0.5)
x
```

```
[1] 2.0 2.5 3.0 3.5 4.0 4.5 5.0 5.5 6.0
```

Sometimes it is necessary to have repeated values for which we use the function rep()

We can repeat 5 3 times as follows

```
rep(5,3)
```

```
[1] 5 5 5
```

Rules for object names in R

1. Object names are case sensitive: object MyObject is not the same as myobject
2. They can contain both numbers or letters but should START with a letter, e.g. 2my-object is not a valid object name.
3. You can use underscores and points. Other special characters should be avoided.
4. Avoid using function names as object names.

Vector Operations

You can add, subtract, multiply and divide vectors.

```
vector1 <- c(1,2,3,4,5)
vector2 <- c(2,3,4,5,6)
# addition
vector1 + vector2
```

```
[1] 3 5 7 9 11
```

```
# subtraction
vector1 - vector2
```

```
[1] -1 -1 -1 -1 -1
```

```
# multiplication
vector1*vector2
```

```
[1] 2 6 12 20 30
```

```
# division
vector1/vector2
```

```
[1] 0.5000000 0.6666667 0.7500000 0.8000000 0.8333333
```